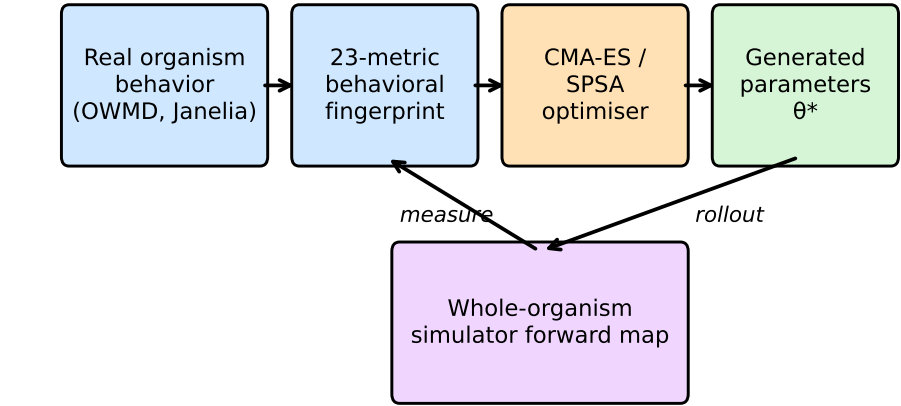


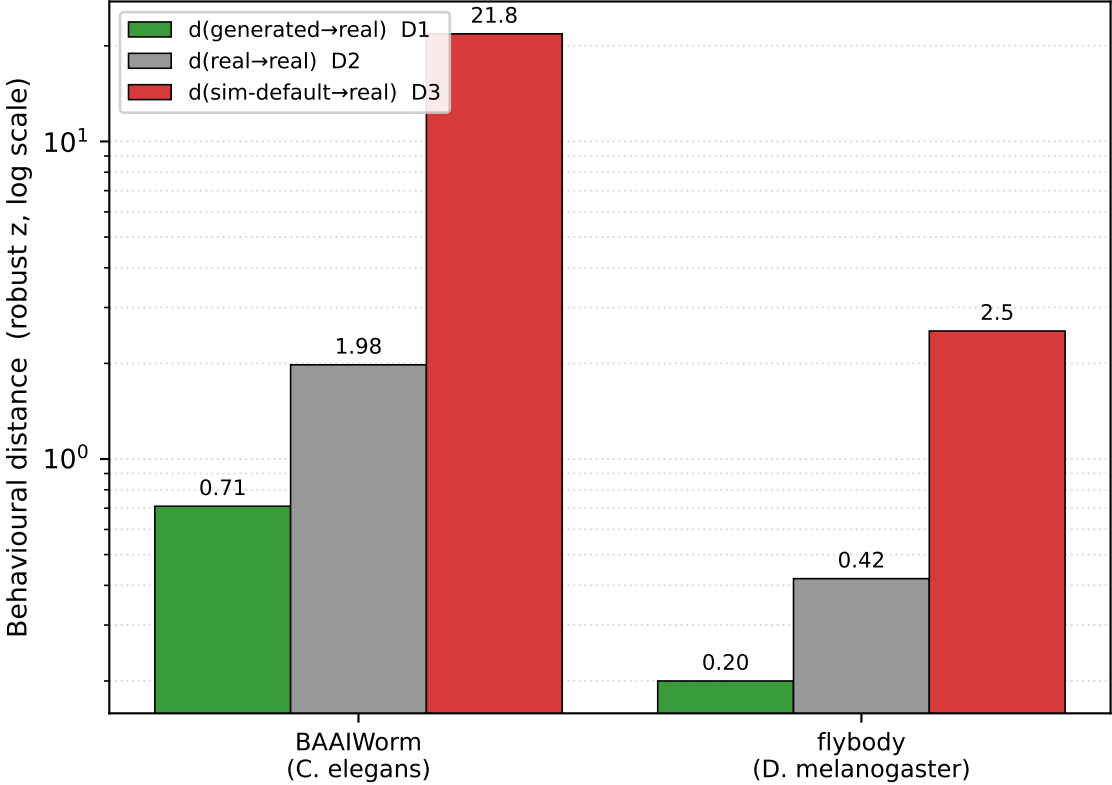
Figure 1 — Behavior-grounded parameter generation (PGB): framework, three-distance hierarchy, falsifiability, mutant differential

A. Behavior-grounded parameter generation (PGB)

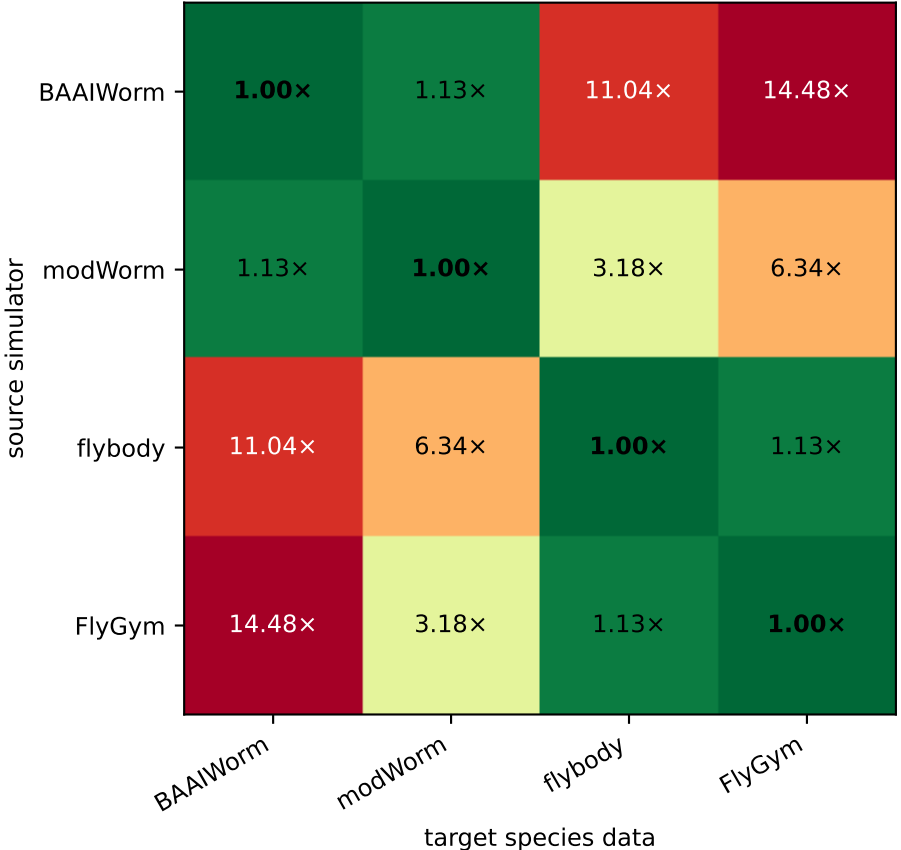
Behaviour-only signal — no prior parameter values used



B. Three-distance hierarchy: $D1 < D2 \ll D3$



C. 4×4 cross-species impossibility (mean 7.57×)



D. Mutant differential (5/4/4 channels, $p < 0.001$)

